Represent tokens in context

AICE3002 / AICE6001

VLC = Vision Learning Control

Tokenisation and Positional Representation

From discrete symbols to contextual representations

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Models operate on representations, not raw text

Words are an unstable unit of representation

Unknown words expose the vocabulary problem

Subwords balance coverage and sequence length

Byte-pair encoding builds a vocabulary by merging

Token IDs become learned vectors

Order must be represented explicitly

Fixed and learned positional representations

Position interacts with attention

Context changes what a token represents

From static embeddings to contextual representations